Check for updates

The Effect of Land Use Restrictions Protecting Endangered Species on Agricultural Land Values

Richard T. Melstrom

This article examines the effect of U.S. Endangered Species Act (ESA) regulations on agricultural values. Agricultural development is an important contributor to habitat and biodiversity loss in the United States. The ESA attempts to limit this loss by prohibiting habitat destruction on private lands, but this practice is controversial because it places much of the burden for conservation on farmers. I measure the effect of these restrictions on agriculture using a hedonic analysis of county-level agricultural land values, profits, and revenues reported in the last four rounds of the agricultural census. Results provide strong evidence that ESA regulations depress these three economic measures in dryland areas, which includes counties with less than 1% of agricultural land in irrigation. Specifically, I find that farmland value and profit decline 4% after listing on average at the county level in dryland areas with protected habitat. There is no evidence that values are affected in irrigated counties.

Key words: Environmental policy, land markets, property values.

Agriculture is a significant source of externalities in the United States. Damage estimates range between $8.0 and $23.7 billion annually (in 2018$) or about 6%–17% of industry output, mostly from impacts to land and soil resources (Tegtmeier and Duffy 2004). These impacts are extensive because farmers control over 40% of U.S. land, including much of the country's prairies, forests, and wetlands (NASS 2014). In fact, damages from agriculture will likely exceed the losses from climate change over the next half century (Srinivasan et al. 2008). Environmental damages are increasing because farmland conservation practices are not keeping pace with agricultural intensification (Tilman et al. 2001; Tscharntke et al. 2005; Swinton et al. 2015).

The Endangered Species Act (ESA) is a federal law that attempts to address some of the externalities from agriculture. The ESA prohibits damaging the habitat of threatened and endangered species. Compliance is mandatory, and there are no loopholes for agricultural operations, unlike other federal environmental laws such as the Clean Water Act and the Clean Air Act.[1] This is because managing agricultural practices is crucial to the recovery of imperiled species (Wilcove et al. 1998). ESA regulations can restrict producers' opportunities to use land in ways that harm an endangered species, which places much of the burden of biodiversity preservation on agricultural landowners because most species rely on private land and thus farmland for habitat (GAO 1994). This has led to fears that ESA regulations "devastate" farmers and ranchers, and motivated farm and political organizations to oppose using the law to protect new species (Fallin, Barrasso, and Foxx 2016; AFBF 2018). Recently, regulators have changed their policies to allow documenting the financial burden of protecting

Richard T. Melstrom is at the Institute of Environmental Sustainability, Loyola University Chicago, Chicago, Illinois. The author would like to thank Leah Vasarhelyi for valuable research assistance.
Correspondence to be sent to: rmelstrom@luc.com

[1]The Clean Water Act excludes most agricultural operations by distinguishing between point and nonpoint source discharges. The Clean Air Act regulates farm airborne pollutants but there are *de facto* exemptions for food producers; most agricultural operations are considered minor sources of air pollution not subject to the Act's permit requirements (Copeland 2010).

Amer. J. Agr. Econ. 103(1): 162–184; doi:10.1111/ajae.12127
Published online September 2, 2020
© 2020 Agricultural and Applied Economics Association

species (FWS 2019), which could help renew efforts to pass legislation requiring regulators to avoid significant land use restrictions unless compensation is offered (Innes, Polasky, and Tschirhart 1998).

In this article, I examine whether ESA land use restrictions reduce agricultural productivity. I focus on the ESA as a key agri-environmental law for two reasons. First, the performance of the ESA and food production are inextricably linked through land use, and much of the controversy around the law centers on its treatment of farmers. Prior research on the ESA focuses almost exclusively on effects in local housing and natural resource markets rather than agriculture (Meyer 1997; List, Margolis, and Osgood 2006; Zabel and Paterson 2006; Greenstone and Gayer 2009; Bošković and Nøstbakken 2017; Melstrom 2017), although there are exceptions (Malone and Melstrom 2020). In contrast, economists are increasingly studying the implications of other environmental policies on agriculture, for example, wetlands conservation and Clean Air Act regulations (Sunding and Zilberman 2002; Njuki and Bravo-Ureta 2015; Zhang 2017). Second, agricultural lands are especially susceptible to ESA regulations because they overlap with endangered species habitat to a greater extent than other land uses, and the restrictions are believed to be particularly onerous (Goodstein 1999).

This study uses a hedonic approach to measure the effects of ESA land use restrictions on agriculture. So far, evidence on the effects of the law in other markets has been mixed, with some studies finding negative effects (Zhang 2004; List et al. 2006) and others no statistically distinguishable effect (Greenstone and Gayer 2009; Wietelman and Melstrom 2017). I extend this research by examining how much land use restrictions have affected agricultural land values, profits, and revenues nationally. The empirical analysis uses the county year as the observational unit, with data drawn from four rounds of the agricultural census (2002 to 2017), a habitat location inventory, and the timing of endangered species listings. The analysis combines a fixed effects panel data estimator with a propensity-score design to identify the effect of regulations. The point estimates unambiguously indicate that listing an endangered species reduces land values in dryland areas, which include counties with less than 1% of their agricultural land irrigated, but not in other areas.

Institutional Information

The ESA makes harming an endangered species illegal, which includes activities that harass, kill, capture, or significantly alter the habitat of a species. This protection extends to habitat on public as well as on private land for listed animals and to federal public land for listed plants. Prohibiting habitat alteration can impose a substantial land use restriction on property owners. The famous snail darter listing in the 1970s halted the construction of a dam to prevent habitat loss.[2] Individual violations can result in civil and criminal penalties, including fines between $25,000 and $50,000, and up to a year in prison. Protecting habitat through land use restrictions makes the ESA a powerful law but fuels conflicts with landowners, which often include farmers and ranchers (Davis 2001; Sheridan 2007). Landowners have argued that prohibiting habitat modification on private property is a compensable take under the Fifth Amendment (Meltz 2009).[3]

The Secretary of the Interior maintains a list of species protected under the ESA. The Fish and Wildlife Service (FWS), under the Department of the Interior, shares responsibility with the National Marine Fisheries Service (NMFS) for listing species. The ESA requires the Secretary to list species at risk of extinction based on the best scientific data available. Species are listed under the ESA as either threatened or endangered, although for simplicity I refer hereafter to listed species in either category as endangered. One of the key ways the ESA protects listed species is by protecting habitat. The FWS publishes the approximate habitat range for each species on its website and is required by law to designate critical habitat areas essential for conservation. Economists have been drawn to the economic consequences of designating critical habitat because it is the only part of the law that allows considering costs and benefits (Zabel and Paterson 2006). Critical habitat areas require federal agencies consult with the FWS about "any agency action which is likely to jeopardize the continued existence of any species … or result in the destruction

[2]The species was listed because its range was restricted to a single section of a stream. The ESA land use restrictions prohibiting construction of the dam were upheld by the Supreme Court, but Congress later passed a special exemption so the dam could be built.

[3]The Fifth Amendment of the U.S. Constitution says "[n]or shall private property be taken for public use, without just compensation."

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

or adverse modification of critical habitat" (§7.a.4). The incremental costs of critical habitat over other provisions of the ESA can affect which areas are excluded from critical habitat (Plantinga, Helvoigt, and Walker 2014). However, the ESA prohibits altering any occupied habitat regardless (Lueck and Michael 2003), which is important because most endangered species have not been assigned critical habitat. This article examines the effect of the ESA wherever an endangered species is present. The Appendix provides additional information about listing status and the extent of ESA regulations.

Besides landowners, regulators and policy makers are interested in the effects of land use restrictions on private property. Several ESA reauthorization bills have proposed full compensation in the event that regulations reduce property values below a threshold (Michael 2000). For example, the 2005 reauthorization proposed requiring the Secretary of the Interior to award aid to landowners prevented from using their property due to the harm prohibition. None of these proposals have become law. However, there remains substantial pressure on policy makers to respond to landowner concerns by reforming the ESA. H.R. 717 in 2017 proposed allowing the FWS and NMFS to consider economic costs when deciding whether to list a species, including effects on private property and reductions in land value. Furthermore, in 2019 the FWS implemented several rule changes affecting how it interprets and implements the ESA. One of these changes removed regulatory language barring reference to economic impacts in the listing determination. Future listing considerations may therefore report the economic costs of listing for the sake of public awareness, although this information is not supposed to affect the FWS's decision (FWS 2019).

Theoretical Effects on Land Value, Profit and Revenue

I measure the effects of ESA regulations on agricultural land values, farm profits, and farm revenues using a hedonic model in which the number of endangered species is a farmland attribute. The model assumes per-acre profit is a function of harvest h, commodity price p, and rental rate r. Farmers choose h with convex costs $c(h, n, x)$, where n is the number of endangered species and x is a set of operator characteristics and other land attributes. Farm profits per acre are given by

$$(1) \quad \pi = ph - c(h,n,x) - r.$$

The first order condition for maximizing profits is

$$(2) \quad \pi = p - c_h = 0.$$

Let h^* satisfy Equation (2). In equilibrium, farmers bid up r until it just equals their willingness to pay to farm the land. Rent is therefore determined by the zero profit condition,

$$(3) \quad r(n,x) = ph^* - c\left(h^*,n,x\right).$$

The value of land equals the discounted sum of rents,

$$(4) \quad v = \theta r(n,x),$$

where θ is the capitalization ratio.

The effect of n on land prices will be negative if protecting endangered species increases production costs, which implies that $c_n > 0$. ESA regulations can affect these costs by compelling a farmer to change their combination of inputs, such as pesticides and herbicides, to avoid killing endangered wildlife; or regulations can compel a farmer to use land less extensively so as not to disturb habitat. Faced with an increase in the number of endangered species, the change in the value of land is therefore

$$(5) \quad \frac{\partial v}{\partial n} = \theta\left[(p - c_h)\frac{\partial h^*}{\partial n} - c_n\right] < 0.$$

The effect of ESA regulations on output and revenue may also be negative. Differentiating equation (2) with respect to n,

$$(6) \quad -c_{hh}\frac{\partial h}{\partial n} - c_{hn} = 0,$$

which indicates that $\frac{\partial h}{\partial n} = -c_{hn}/c_{hh}$. Both c_{hn} and c_{hh} can be positive if costs are based on Cobb-Douglass technology, implying that harvest and therefore revenue decreases in the number of endangered species. Thus, if ESA regulations burden farms with additional costs, as is claimed, increasing numbers of endangered species should be associated with lower land value, profit, and revenue.

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Nevertheless, other land use activities could temper or shift the effect of ESA regulations in the other direction. Landowners could preemptively manage their land in a way to avoid ESA regulations, by either destroying or protecting habitat. Preemptive habitat destruction has been the focus of some research in economics (Lueck and Michael 2003; Zhang 2004), but there are also high-profile cases of preemptive conservation intended to avoid listing, such as the efforts to protect the greater sage grouse and the lesser prairie chicken. These activities would reduce the expected costs of protecting an endangered species if listing were to occur, and thus mitigate the effect of ESA regulations on agricultural values. There is also an important relationship among residential land, location preferences, and endangered species. ESA regulations near urban areas can increase the price of land adjacent to habitat by inadvertently functioning as a growth boundary (Quigley and Swoboda 2007). Moreover, some households may be willing to pay to live near unfragmented natural areas, which could also serve to increase land values around habitat (Geoghegan, Wainger, and Bockstael 1997; Ready and Abdalla 2005). The empirical strategy described below controls for these potential confounders by excluding agricultural land near urban areas and by combining a matching design with a fixed effects panel data estimator that includes a variety of land use factors.

Fixed Effects Panel Data Estimator and Matching

The basic hedonic formula I use features a full set of location and time effects,

$$(7) \qquad v_{ct} = \delta n_{ct} + \boldsymbol{X}_{ct}\boldsymbol{\beta} + \sum_i \boldsymbol{\theta}_i f_i(\boldsymbol{W}_{ict}) + \alpha_c + \gamma_{ct} + u_{ct}$$

where v_{ct} is either the value of agricultural land, profit, or revenue per acre; and n is the number of endangered species with habitat in county c and year t. $\boldsymbol{X}_{ct}$ includes operator characteristics and local farm attributes. $\boldsymbol{W}_{ict}$ are annual realizations of weather for climate variable i that are known to affect the profitability and thus the value of farmland. Specifically, f_i includes measures of spring, summer, fall, and winter temperature and precipitation in separate quadratic terms (Mendelsohn, Nordhaus, and Shaw 1994; Mendelsohn and Dinar 2003; Deschenes and Greenstone 2007). I control for the influence of time-constant land attributes through county fixed effects α_c. The effects of common macroeconomic shocks, such as changes in commodity prices and region-specific unobservables, are absorbed through census division-by-year effects γ_{ct}.

Embedded in the fixed effects panel data model are several crucial assumptions. First, counties exhibit the same response to common shocks. This assumption would be violated if, for example, an increase in commodity prices exposed some counties to ESA regulations more than others due to their habitat characteristics. Second, the treatment effect is additive and homogenous, which would, for example, be violated if ESA regulations reduced values more on average in counties with more productive land. Third, the model assumes the functional form is linear. Failing these assumptions can create significant bias in nonexperimental studies that rely on fixed effects to identify the effect of interest (Ferraro and Miranda 2017).

This study attempts to overcome these problems by processing the data in two significant ways. First, I use matching to develop a group of unregulated counties that is observationally similar to the group affected by ESA regulations, just as there would be in a randomized controlled trial. This involves selecting counties that have never had an endangered species but appear similar to counties with endangered species in terms of important covariates. Research shows that fixed effects estimators can generate much more accurate estimates when combined with matching (Imbens and Wooldridge 2009; Ferraro and Miranda 2017). Second, I estimate the model separately for irrigated and dryland counties. Research shows that these areas can exhibit significantly different responses to the same variables in hedonic models (Schlenker, Hanemann, and Fisher 2005).

To match the data, I use nearest-neighbor matching on the estimated propensity score (Rosenbaum and Rubin 1985). The propensity score is the conditional probability that a county contains at least one endangered species in the study period. Counties with and without endangered species that have the same propensity score will have the same characteristics, which increases the credibility of the assumption that regulated and unregulated counties respond the same way on average to common shocks, and the assumption that the effect of ESA regulations is the same

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

on average across counties. Matching also reduces concerns about the linear functional form (Ferraro and Miranda 2017). To generate the propensity scores, I considered a county to be affected by ESA regulations if it contained at least one endangered species at any time in the study period. The covariates used to estimate the propensity score include measures potentially correlated with endangered species and habitat, including land value, profit, revenue, and population density. I applied the matching with a caliper equal to 0.2 times the standard deviation of the propensity score (Wang et al. 2013). The caliper eliminates counties that do not have a good match and ensures the similarity between the regulated and unregulated groups. Counties that do not match are dropped.

The irrigated and dryland groups include counties with more or less than 1% of farmland irrigated, respectively.[4] Converting land to agriculture damages native vegetation and habitat; however, irrigation can affect the suitability of habitat in systematically different ways than dryland farming, for example by supplying water, which is critical to the survival of every organism. Furthermore, estimating separate fixed effects models is important because prior research finds that dryland and irrigated counties cannot be pooled without bias (Schlenker et al. 2005). Adding a single shift parameter in a pooled model is insufficient because irrigation technology can fundamentally change how environmental variables affect production, leading to different model coefficients.

I weight observations by farmland acres, which includes cropland and pastureland. These are the same weights one would use to estimate the average value of agricultural land from the county summaries published by the U.S. Census of Agriculture. Other weights, including cropland and crop revenue weights, have been used in prior research on the hedonics of farmland values (Darwin 1999; Schlenker et al. 2005). Below, I probe the sensitivity of the estimates to different weighting strategies.

I allow correlation between farmland values over time by clustering standard errors on the U.S. Department of Agriculture's (USDA) crop reporting districts. The USDA divides the country into several hundred crop districts made up of counties with similar crop types and weather (Guttman and Quayle 1996). This clustering allows for spatial correlation of counties in the same districts (Cameron and Miller 2015).

Data

This analysis draws largely on two datasets. The first is the Census of Agriculture's county-level summaries from the 2002, 2007, 2012, and 2017 censuses. The variable of primary interest, agricultural land value, includes the market value of land and buildings. These data are per-acre, county-level averages from self-reported estimates (Plantinga and Miller 2001). In Appendix I show that these values closely align with the prices from actual land sales. I also collected census data on farm revenue and production expenses. Farm revenue is defined as the market value of products sold, which measures receipts before taxes and excludes any payments from federal farm programs. Farm production expenses include labor costs, paid rents, property taxes, and interest on loans. I constructed the farm profit variable by subtracting expenses from revenues per acre (Deschenes and Greenstone 2007).

The agricultural census also contains data on several key covariates. For operator characteristics I include average age, age squared and the percent of farmland managed by female producers. To control for land use, I include three variables measuring the percent of farmland in crops, irrigation, and conservation easements. Land values should be higher in counties with more cropland, especially irrigated cropland. The conservation easement variable includes land enrolled in federal rental payment programs that remove land from production, including the Conservation Reserve Program, reported by the Census of Agriculture. The easement variable also includes land enrolled in public and private easements reported through the National Conservation Easement Database.

Table 1 presents summary statistics for the economic and land use variables used in the analysis. In addition to agricultural data, I collected median household income and county population density from the U.S. Census Bureau's Small Area Income and Poverty Estimates program. All values are converted to current 2017 dollars using the consumer

[4]This divides the sample between 947 irrigated and 1,269 dryland counties after matching. I find coefficient signs and significance levels are insensitive to different cutoff levels (e.g. 5%).

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 1. Summary Statistics

A. All counties

Variable	Mean	St. dev.	Min	Max
Farm acres (1000s)	313.31	377.97	1.45	4595.06
Land value	3134.48	2089.66	102.73	44154.33
Farm revenues	496.25	604.07	2.75	8934.82
Farm expenses	367.03	428.18	2.06	6825.15
Farm profit	85.46	166.12	−392.08	2872.41
Operator age	57.40	2.47	45.10	70.00
Operator female	11.67	10.51	0.01	94.21
Median income (1000s)	47.02	10.03	20.47	102.73
Population/sqmi	51.58	49.82	0.26	362.59
Crops (%)	49.32	26.02	0.00	98.65
Irrigation (%)	5.02	11.16	0.00	92.69
Conservation (%)	3.52	4.09	0.00	36.37
Presence of ES	0.73	0.44	0.00	1.00
Number of ES	2.07	3.09	0.00	34.00

B. Type of county	Irrigated counties		Dryland counties	
Variable	Mean	St. dev.	Mean	St. dev.
Farm acres (1000s)	366.45	412.15	273.66	345.05
Land value	3156.12	2460.72	3118.33	1762.66
Farm revenues	576.61	663.74	436.27	547.82
Farm expenses	427.22	486.52	322.06	372.52
Farm profit	98.47	169.27	75.73	163.07
Operator age	57.44	2.46	57.38	2.48
Operator female	11.23	10.72	12.00	10.34
Median income (1000s)	47.69	10.35	46.52	9.76
Population/sqmi	49.09	52.87	53.45	47.34
Crops (%)	51.11	26.07	47.98	25.91
Irrigation (%)	11.39	14.84	0.27	0.34
Conservation (%)	3.87	4.20	3.27	3.99
Presence of ES	0.75	0.43	0.72	0.45
Number of ES	2.42	3.71	1.82	2.49

price index. To help avoid confounding the effects of the ESA on agriculture with residential and industrial development, I exclude counties with more than 10% of land classified as urban. I also exclude counties in Alaska and Hawaii or those with missing/zero values for agricultural land, land values, and farm revenues in any year. The final panel includes 2,216 rural counties, about three-quarter of all U.S. counties, after matching. On average, a farm is worth $3,134 per acre, with revenue and profit of $496 and $85 per acre, respectively. Farmers are older and more likely to be male than non-farmers. Nearly half of farmland is in crops, but only about 5% is irrigated.

The second dataset is NatureServe's 2016 biodiversity inventory of the United States. This inventory includes the location of every listed, endangered species. There are over 160,000 species-county records. Each species record includes ESA status and an element occurrence score. I consider a species to be present in a county if the occurence score indicates that a field survey has confirmed the presence of the species.[5] The Appendix provides additional information about these occurrence scores and an alternative source of habitat data. Because the NatureServe data is not a panel, I assume habitat extant in the reporting year, 2016, was extant in the census years.[6] I assigned an endangered species to a county year if the species was listed and present in the county prior to or in the census year. Figure 1 shows the locations of species

[5]Note that I excluded the bald eagle as an endangered species because it receives special protections under the Bald and Golden Eagle Protection Act.

[6]NatureServe also tracks where species have been recently extirpated. When I estimate the models with the number of listed species, including listings in counties with evidence of extirpation, the results are very similar to those reported in the paper.

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

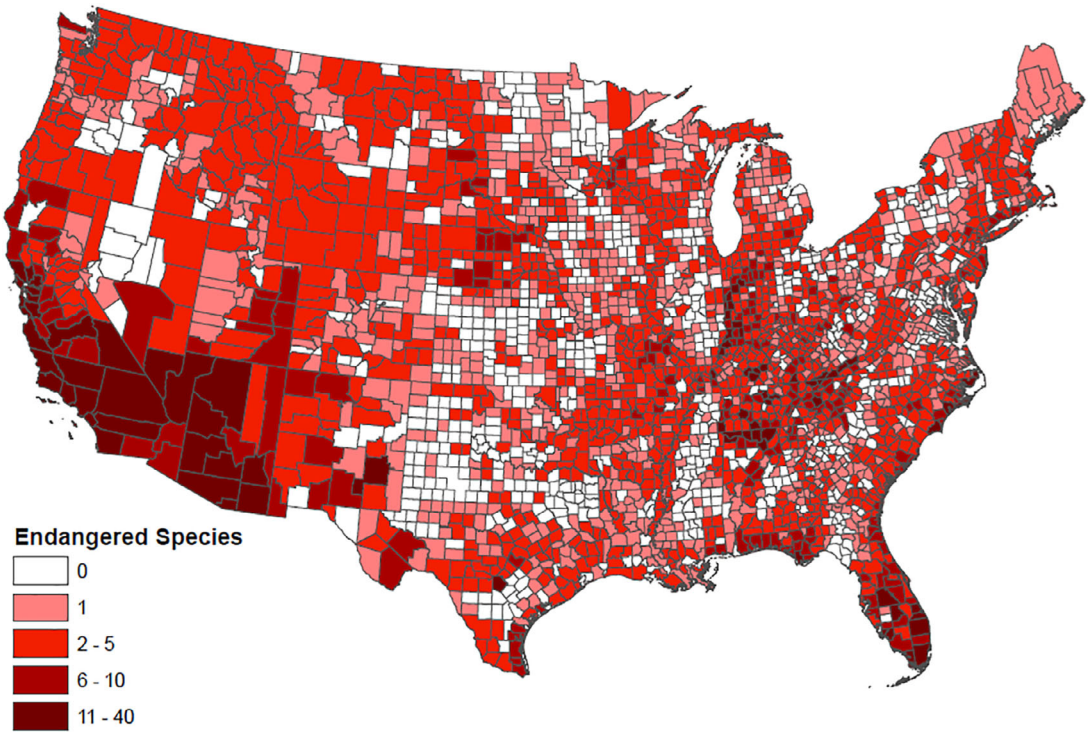

Figure 1. Distribution of endangered species [Color figure can be viewed at wileyonlinelibrary.com]

listed as of 2017. In fact, listed species are quite widespread, with over half of counties containing one or more, and 39% containing two or more. The map also shows that many listings are located in regional "hotspots" with more than ten endangered species (about 5% of counties), which tend to be located near the coast or major river systems.

I also collected land capability class data recorded by the USDA's Natural Resource Conservation Service and weather data from NOAA's National Climate Data Center. Land capability class indicates the suitability of soils for farming. The model includes variables for the percent of farmland in each class. The weather data include monthly temperature and precipitation. I generated quadratics of temperature and precipitation for January, April, July, and October, following the convention in the literature (Mendelsohn et al. 1994; Mendelsohn and Dinar 2003; Deschenes and Greenstone 2007).

Table 2 shows how propensity score matching increases the similarities between counties with and without endangered species. I estimated the propensity scores using the 2002 values of the covariates listed in the table. For each variable, the table presents group averages before and after matching. Before matching, the two groups are substantially different across several key variables, including land value and population density. After matching, the two groups are much more similar in terms of these variables. The last column reports the standardized difference between matched counties with and without endangered species, which are below tolerable limits (Rosenbaum and Rubin 1985). The number of counties with endangered species is the same pre- and post matching because the caliper does not eliminate any matches.[7]

Results and Discussion

Primary Results

Table 3 presents the regression estimates when dryland and irrigated counties are pooled together. The coefficient on the

[7]I find the results are largely insensitive to the caliper distance because there are few extreme propensity scores differences. For example, a caliper one-tenth the distance of the baseline caliper trims away only seven matches (56 observations).

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 2. Variables Means across Counties With and Without Listed Species

Variable	Counties with no ES – All	Counties with ES – All	Counties with no ES – Matched	Counties with listed ES – Matched	Standardized difference
Land value	2566.94	3258.80	3256.35	3258.80	0.12
Profit	76.05	88.24	86.91	88.24	0.85
Revenue	449.87	516.07	502.87	516.07	2.27
Population/ sqmi	37.33	54.53	55.26	54.53	−1.57
Crops	52.15	49.22	48.04	49.22	4.48
Irrigation	6.60	5.10	4.07	5.10	8.10
Conservation	4.29	3.49	3.18	3.49	7.01
Counties	871	1628	588	1628	

The standardized difference is calculated as $100 \times (x_T - x_M)/\sqrt{(s_T^2 + s_R^2)/2)}$ where x_T and x_M are the means across matched counties with and without endangered species (ES), respectively, and s_T and s_R are the standard deviations in matched counties with ES and all counties without ES, respectively. Rosenbaum and Rubin (1985) suggest that bias is problematic when a standardized difference exceeds 20.

number of endangered species in column (1) indicates that listing reduces land values \$81/acre ($p < 0.05$) on average. Many of the other coefficients in the model have the expected signs and significance levels, for example the estimates indicate that land values are higher on average in areas with more cropland and irrigation. The effect of the number of endangered species is negative but not statistically significant in the profit and revenue models.

Columns (4), (5), and (6) in Table 3 separate the number of endangered species into two variables, one of which numbers endangered species in counties with more than one and the other which serves as an indicator for counties with a single endangered species. The distinction between zero, one and more than one listing could be important if regulations have a diminishing marginal effect in the number of endangered species. The estimates in column (4) indicate that the first and additional listings reduce land values \$167/acre ($p < 0.10$) and \$61/acre ($p < 0.10$), respectively; however, the difference is not statistically significant ($p = 0.24$). The same coefficient is negative in the profit and revenue models in columns (5) and (6), respectively, but imprecisely estimated. Figure 2 shows the effect on land values when I include separate indicators for one, two, and so on for nine endangered species, with the final category measuring the number of endangered species in counties with ten or more.[8] The figure shows quite clearly how land values change with respect to different numbers of endangered species and thus the intensity of ESA regulations. There is no appearance of a diminishing marginal effect until the number of endangered species reaches seven.

Table 4 reveals important differences between dryland and irrigated counties. Columns (1), (2), and (3) provide little evidence that ESA regulations affect farms in irrigated counties. With respect to the number of endangered species, not one coefficient is significantly negative in these models. In contrast, columns (4), (5) and (6) provide strong evidence that regulations affect farms in dryland counties. The coefficients in these models imply that listing results in a loss of \$131/acre (4%) in farmland value, \$4/acre (5%) in farm profit and \$10/acre (2%) in farm revenue. The effect on land value is highly statistically significant ($p < 0.01$), and the effects on profit and revenue just reach a threshold of significance ($p < 0.10$).

The coefficients in table 4 provide strong indication that ESA regulations affect farm values in dryland areas but may not affect values in irrigated areas. To be clear, there may be effects in irrigated areas—for example, we cannot reject the hypothesis that listing reduces land values in irrigated counties by \$100/acre—but we also cannot reject the possibility of no effect. The remainder of the analysis therefore deals exclusively with the effects in dryland counties.[9]

Table 5 presents the estimates when the number of endangered species is separated

[8]Other coefficients not shown for brevity but available upon request.

[9]Furthermore, Schlenker et al. (2005) argue that hedonic farmland estimates for irrigated areas may be very unreliable without measures of the availability of irrigation water, which do not exist in national-level data.

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 3. Coefficient Estimates in Farmland Value, Profit, and Revenue Models with Irrigated and Dryland Counties Pooled

	Standard models			Adding an indicator for one ES		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−81.016**	−1.069	−2.681	−61.089*	−0.307	−1.190
	(38.807)	(2.135)	(5.250)	(33.172)	(1.903)	(4.535)
One ES				−167.245*	−8.193*	−7.817
				(93.907)	(4.912)	(11.307)
Operator age	1231.283***	113.686***	374.660***	1233.113***	113.844***	374.829***
	(363.762)	(24.237)	(69.353)	(364.008)	(24.236)	(69.416)
Operator age^2	−10.216***	−0.966***	−3.192***	−10.232***	−0.967***	−3.193***
	(3.094)	(0.207)	(0.594)	(3.096)	(0.207)	(0.594)
Operator female	−3.111	−0.061	0.041	−3.226*	−0.067	0.035
	(1.888)	(0.126)	(0.337)	(1.900)	(0.127)	(0.338)
Income	−5.217	−0.967***	−2.957***	−5.399	−0.980***	−2.966***
	(7.011)	(0.324)	(0.823)	(7.021)	(0.325)	(0.824)
Population/sqmi	26.843***	−0.467**	2.189***	26.781***	−0.469**	2.186***
	(4.577)	(0.192)	(0.801)	(4.562)	(0.192)	(0.800)
Crops	20.686***	2.149***	6.243***	20.697***	2.147***	6.243***
	(5.930)	(0.348)	(1.076)	(5.935)	(0.348)	(1.076)
Irrigation	52.433***	0.851	10.577***	52.422***	0.854	10.579***
	(14.876)	(0.930)	(2.980)	(14.894)	(0.929)	(2.980)
Conservation	30.726***	−0.630	−3.517**	30.352***	−0.642	−3.535**
	(8.526)	(0.656)	(1.440)	(8.489)	(0.658)	(1.441)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	8864	8856	8864	8864	8856	8864

Standard errors in parentheses. Standard errors are clustered on crop districts, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

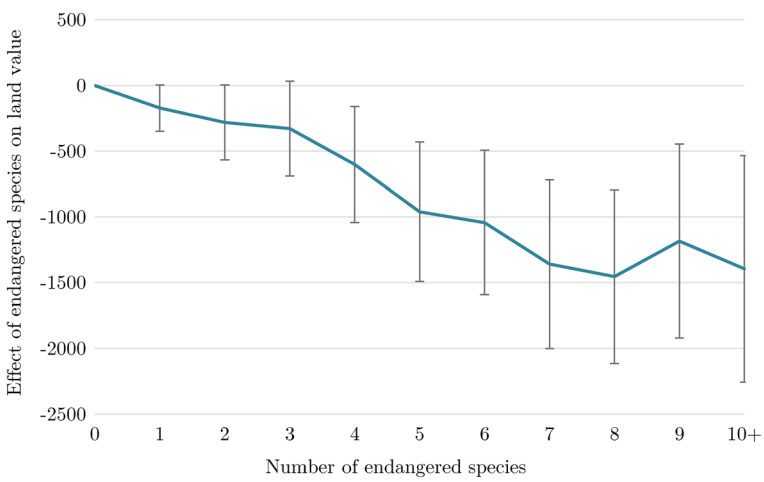

Figure 2. The effect of the number of endangered species on agricultural land values.

Note: The estimates are coefficients on indicators for different numbers of endangered species in a county. The vertical bars show 95% confidence intervals. [Color figure can be viewed at wileyonlinelibrary.com]

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 4. Coefficient Estimates Separating Irrigated versus Dryland Counties

	Irrigated counties			Dryland counties		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−25.396	1.782	−1.368	−130.721***	−4.383*	−9.899*
	(58.341)	(2.763)	(8.378)	(36.042)	(2.435)	(5.171)
Operator age	383.705	61.548**	252.864***	2057.703***	164.484***	481.738***
	(393.265)	(25.491)	(76.324)	(465.996)	(36.057)	(85.838)
Operator age^2	−3.299	−0.522**	−2.168***	−17.249***	−1.404***	−4.111***
	(3.355)	(0.216)	(0.655)	(3.982)	(0.311)	(0.735)
Operator female	−3.355	−0.094	0.099	−4.647	−0.002	−0.037
	(2.362)	(0.143)	(0.497)	(3.246)	(0.207)	(0.449)
Income	6.106	−0.608*	−2.133*	−8.148	−1.064**	−3.042***
	(8.793)	(0.352)	(1.098)	(8.039)	(0.483)	(1.082)
Population/sqmi	32.006***	−0.320	3.997***	16.861***	−0.668***	−0.190
	(6.870)	(0.302)	(1.274)	(4.241)	(0.249)	(0.561)
Crops	17.567**	1.138***	6.275***	22.650***	2.742***	5.967***
	(7.621)	(0.426)	(1.597)	(6.402)	(0.474)	(1.098)
Irrigation	64.871***	1.477	11.716***	154.673***	10.445**	28.907***
	(14.301)	(0.939)	(2.766)	(58.706)	(4.902)	(10.765)
Conservation	18.767**	0.090	−2.365	50.183***	−1.335	−4.574**
	(9.376)	(0.854)	(1.637)	(12.678)	(0.915)	(2.268)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	3788	3787	3788	5076	5069	5076

Standard errors in parentheses. Standard errors are clustered on crop districts, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

between animals and plants. This disaggregation provides a test of whether the effect of listing endangered species varies with the severity of land use restrictions, because the ESA's prohibition on harming listed species on private land does not include plants. This means landowners experience land use restrictions based principally on the extent of endangered animals. Columns (1), (2), and (3) present separate coefficients for the number of endangered animals and plants. The effect of endangered animals is negative and highly significant in all models ($p < 0.05$). The coefficients imply that listing animals in dryland counties results in a loss of \$133/acre (4%) in farmland value, \$5/acre (6%) in farm profit, and \$12/acre (2%) in farm revenue. The effect of endangered plants is not significantly different from zero except, curiously, in the revenue model, in which the effect is actually positive ($p < 0.05$). Overall, these estimates provide strong evidence that ESA regulations protecting endangered animals but not plants reduce farm productivity.

I next examine an important channel through which land use restrictions could be enforced: designating critical habitat. Industry advocates have long argued that designating critical habitat imposes substantial costs on landowners (Turner and McGrath 2013). Designation affects activities that involve federal activity in critical habitat, which could affect farmers' access to federal technical and financial assistance, such as the Environmental Quality Incentives Program (EQIP), in critical habitat. Recall, however, that regardless of critical habitat designations, farmers are prohibited from actions that harm a listed species. To measure the cost associated with critical habitat designations, I separated the number of endangered species between those with and without critical habitat. The estimates are presented in columns (4), (5), and (6) of Table 5. The coefficient on the number of endangered species without critical habitat is negative and statistically significant in both the land value and revenue models ($p < 0.05$). In contrast, the coefficient on endangered species with critical habitat is not significant in any model. Thus, there is little evidence that listing species with critical habitat affects farmers. In fact, the evidence suggests that

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 5. Coefficient Estimates for Dryland Counties with Separate Counts for Species Type and Critical Habitat

	Listed animals vs. plants			Listings with vs. without critical habitat		
	(1)	(2)	(3)	(4)	(5)	(6)
	Land value	Profit	Revenue	Land vValue	Profit	Revenue
Number of animal ES	−133.335***	−5.057**	−12.061**			
	(38.055)	(2.508)*	(5.357)			
Number of plant ES	−77.996	9.193	33.696**			
	(60.191)	(9.221)	(15.631)			
Number of ES w/o CH				−210.561***	−3.320	−15.237**
				(45.011)	(3.012)	(6.407)
Number of ES w/ CH				48.187	−6.766	2.062
				(55.255)	(4.427)	(8.508)
Operator age	2059.656***	164.987***	483.352***	2056.426***	164.501***	481.652***
	(466.257)	(36.095)	(85.931)	(465.722)	(36.086)	(85.786)
Operator age^2	−17.267***	−1.408***	−4.126***	−17.239***	−1.404***	−4.110***
	(3.984)	(0.311)	(0.736)	(3.979)	(0.311)	(0.735)
Operator female	−4.792	−0.039	−0.157	−3.310	−0.020	0.052
	(3.289)	(0.207)	(0.442)	(3.310)	(0.204)	(0.450)
Income	−8.187	−1.074**	−3.073***	−8.550	−1.058**	−3.068***
	(8.030)	(0.483)	(1.081)	(8.039)	(0.484)	(1.086)
Population/sqmi	16.860***	−0.668***	−0.191	16.916***	−0.668***	−0.186
	(4.239)	(0.249)	(0.562)	(4.298)	(0.248)	(0.562)
Crops	22.632***	2.738***	5.953***	22.357***	2.746***	5.948***
	(6.396)	(0.475)	(1.096)	(6.336)	(0.475)	(1.093)
Irrigation	154.339***	10.359**	28.631***	158.672***	10.391**	29.175***
	(58.700)	(4.911)	(10.781)	(58.456)	(4.915)	(10.742)
Conservation	50.174***	−1.337	−4.581**	51.602***	−1.354	−4.479**
	(12.672)	(0.913)	(2.268)	(12.721)	(0.923)	(2.269)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	5076	5069	5076	5076	5069	5076

Standard errors in parentheses. Standard errors are clustered on crop districts, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

listing without critical habitat is more effective. One potential explanation is that landowners with critical habitat remove habitat from their lands in order to avoid regulations when a species is listed. Prior research finds that FWS proposals to designate critical habitat can serve as an advanced warning that provokes preemptive habitat destruction (List et al. 2006). There is also some evidence that agricultural lands with critical habitat develop at faster rates than other agricultural lands harboring endangered species (Nelson et al. 2017).[10] Although more research is needed to examine how farmers respond to critical habitat, farmers using the information in FWS proposals to remove habitat could explain the differences in Table 5.[11]

[10]Nelson et al. (2017) show that critical habitat areas established between 1999 and 2003 subsequently experienced more agricultural development relative to control areas harboring endangered species. However, the size and significance of the difference depends on the matching method. The difference is the same for no matching and nearest neighbor matching (although only the former is significant), but notably smaller and not significant for Mahalanobis matching.

[11]Another explanation is that endangered species with critical habitat rarely occur on agricultural land. In fact, less than 10% of critical habitat overlaps with agricultural land cover (Nelson et al. 2017). It is not clear if regulators have intentionally avoided designating critical habitat for species near agricultural land, which could create problems with endogeneity. It is true that the FWS has displayed a general reluctance to designate critical habitat, however since the late 1990s lawsuits and court orders have helped check this behavior (Nelson et al. 2017).

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Robustness Checks

I next conduct a series of robustness checks to probe the sensitivity of the results to characteristics of the study design. All regressions are performed on the panel of dryland counties. Table 6 presents the estimates from two additional sets of regressions. Columns (1), (2), and (3) show the performance of the estimator without propensity score matching; these coefficients come from the standard fixed effects analysis typically used to measure the economic effects of programs and policies. The estimates in these columns indicates that listing reduces land value \$119/acre, profit \$4/acre, and revenue \$10/acre on average in counties with habitat, which are not significantly different from the effects reported in columns (4), (5), and (6) in table 4. The other columns in table 6 present the coefficients when the observations used in the fixed effects model are weighted by the inverse of the propensity score. Combining regression and weighting can be a useful alternative to matching on the propensity score (Imbens and Wooldridge 2009). As with the standard fixed effects estimator, the estimates are in line with those generated from the matched data. Thus, the results are robust to several different identification strategies.

Next, I check the sensitivity of the results to different types of weights. Table 7, columns (1), (2), and (3) show the coefficients when observations are weighted by the amount of cropland, and columns (4), (5), and (6) show the coefficients when the observations are not weighted at all. Weighting by cropland rather than farmland reduces the influence of counties with large amounts of pasture in favor of counties with more cropland and thus more valuable land overall. In these models, listing reduces land value \$189/acre ($p < 0.01$), profit \$5/acre ($p < 0.10$), and revenue \$15/acre ($p < 0.05$). With no weights, the estimates imply that listing reduces land value \$132/acre ($p < 0.01$) and revenue \$12/acre ($p < 0.05$); the effect on profit is not statistically significant.

I find the results are robust to a number of other empirical adjustments, which are presented in the Appendix. Coefficients change very little when counties with large numbers of endangered species are dropped. The coefficients are also qualitatively similar if the model is applied to counties with a relatively even mix of pasture and cropland; although the number of endangered species has a smaller effect on land value than indicated in previous regressions, the effect is still negative ($p < 0.10$), as are the effects on profit ($p < 0.10$) and revenue ($p < 0.05$). I find some evidence that listing aquatic animals reduces farm revenue more than listing nonaquatic animals ($p < 0.05$), although the estimates are relatively imprecise.[12] I also find that the coefficients do not hinge on a single source of habitat data; when I use county-level habitat locations published by the FWS, the estimates are similar to those based on the NatureServe locations.

Next, I explored the sensitivity of the estimates to different varieties of fixed effects. To be clear, the preferred specification in this paper includes census division-by-year effects, which control for unobserved regional price responses and other economic shocks. As shown in the Appendix, replacing these fixed effects with year fixed effects produces generally larger coefficients, while replacing them with state-by-year effects yields smaller coefficients. For example, the model that includes county and year fixed effects indicates that land values decline \$170/acre ($p < 0.01$) after a listing on average, but the same effect is \$64/acre ($p < 0.05$) in the model with county and state-by-year effects. However, including just county and year fixed effects leaves the estimates more susceptible to omitted variables bias. The specification with state-by-year effects is less susceptible to bias from unobserved local shocks than the specification that uses division-by-year effects. However, using state-by-year effects makes the estimates more susceptible to measurement error bias by absorbing too much variation in the agricultural data to allow for meaningful inference (Fisher et al. 2012). Nevertheless, regardless of the specification, the estimates indicate that listing endangered species reduces land values.

Finally, I report the bootstrapped mean and variance of the effect of listing in dryland counties to further demonstrate the stability and precision of the estimates.[13] Ten thousand

[12]The effect of ESA regulations may depend on taxonomic class or individual species due to habitat extent, needs, and so on. Unfortunately, although there are many listings in the study period, there are only a handful within specific classes, including birds and mammals. This is shown in the appendix. Therefore, I do not measure the effect of ESA regulations for smaller taxonomic ranks.

[13]Bootstrapping is often used in the treatment effects literature, and its validity in propensity score methods is an active area of research. Note that Abadie and Imbens (2008) show bootstrapped standard errors for the nearest-neighbor matching estimator can be misleading.

Table 6. Coefficient Estimates for Dryland Counties Using Different Identification Strategies

	No matching			Inverse propensity score weighting		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−119.322***	−4.363*	−10.071**	−144.594***	−6.037**	−12.609**
	(33.941)	(2.227)	(4.624)	(38.034)	(2.544)	(5.305)
Operator age	1826.783***	127.398***	406.069***	1756.061***	125.496***	406.803***
	(360.753)	(24.691)	(64.911)	(336.671)	(24.017)	(63.289)
Operator age^2	−15.337***	−1.083***	−3.461***	−14.640***	−1.057***	−3.453***
	(3.079)	(0.213)	(0.558)	(2.885)	(0.207)	(0.545)
Operator female	−2.318	−0.015	0.007	−1.714	−0.017	−0.168
	(2.610)	(0.176)	(0.347)	(2.966)	(0.239)	(0.465)
Income	−16.772**	−1.089***	−3.492***	−17.298**	−1.311***	−4.223***
	(6.536)	(0.337)	(0.769)	(6.917)	(0.364)	(0.841)
Population/sqmi	18.614***	−0.619***	0.126	22.122***	−0.199	0.771
	(3.519)	(0.223)	(0.465)	(3.411)	(0.293)	(0.639)
Crops	23.030***	2.126***	4.695***	20.154***	1.827***	3.513***
	(6.787)	(0.336)	(0.977)	(7.069)	(0.387)	(1.152)
Irrigation	177.456***	11.718***	32.449***	166.669***	12.917***	33.893***
	(44.440)	(3.448)	(6.959)	(49.920)	(3.974)	(9.552)
Conservation	43.475***	−1.670**	−3.904**	44.909***	−1.572**	−4.417*
	(10.768)	(0.691)	(1.599)	(10.620)	(0.764)	(2.285)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	5548	5541	5548	5548	5541	5548

Standard errors in parentheses. Standard errors are clustered on crop districts, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

bootstrap replications imply that listing in dryland counties reduces land value \$142/acre (s.e. = \$30), profit \$6/acre (s.e. = \$2), and revenue \$13/acre (s.e. = \$4), which are quite similar to the dryland county estimates in Table 4.

Additional Discussion

The estimates shown in table 4 indicate that listing endangered species reduces farm production in dryland areas. The coefficients imply that listing an endangered species reduces land value \$131/acre in counties with habitat; the same effect on profit is \$4/acre. The reduction is proportionately very similar for profit and land value, which is evidence that declines in production are almost fully capitalized into land values. The farmland value:profit ratio at the average in the data is thirty-seven, slightly higher than the average price:rent ratio over the same period nationally (Baker, Boehlje, and Langemeier 2014). Similarly, the value:profit ratio associated with listing an endangered species is about thirty.

The effect on land values estimated here is larger but qualitatively similar to recent research that reports listing endangered animals reduces nearby home values by 2% (Greenstone and Gayer 2009) and rural property values by 2% (Wietelman and Melstrom 2017).[14] In contrast, recent research examining the impacts of the Species at Risk Act in Canada found that producers pay 24% less for oil leases in protected habitat on public land (Bošković and Nøstbakken 2017), substantially greater than the effect of ESA regulations on farm profits and land values on the margin. The difference could be due to differences in U.S. and Canadian endangered species law, monitoring of protected lands, or the overall number of endangered species in an area.

[14]However, these effects were not significantly different from zero. See Table 3 in Greenstone and Gayer (2009) and the description of the results on page 159 in Wietelman and Melstrom (2017).

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table 7. Coefficient Estimates for Dryland Counties under Different Weighting Strategies

	Cropland weights			No farm area weights		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−189.257***	−5.132*	−14.991**	−132.456***	−2.446	−11.895**
	(47.672)	(2.796)	(6.260)	(33.982)	(3.550)	(5.207)
Operator age	2317.476***	207.349***	598.885***	1309.449***	97.488**	346.690***
	(543.192)	(52.573)	(104.197)	(407.642)	(41.833)	(91.951)
Operator age^2	−19.501***	−1.787***	−5.166***	−11.003***	−0.815**	−2.978***
	(4.744)	(0.462)	(0.905)	(3.489)	(0.367)	(0.795)
Operator female	−18.018***	−0.358	−1.063	−6.781**	−0.396	−0.192
	(5.248)	(0.421)	(0.758)	(3.039)	(0.476)	(1.015)
Income	1.262	−1.403*	−2.690	6.459	−0.703	−2.620
	(10.801)	(0.739)	(1.642)	(8.547)	(0.878)	(1.745)
Population/sqmi	15.981***	−0.725*	−0.436	18.504***	−0.375	1.241
	(5.043)	(0.370)	(0.793)	(3.423)	(0.369)	(0.821)
Crops	28.075***	3.655***	8.037***	27.606***	3.299***	7.919***
	(7.021)	(0.515)	(1.080)	(5.653)	(0.698)	(1.222)
Irrigation	193.902***	15.592**	41.800***	104.445*	11.988*	19.774
	(69.499)	(7.795)	(14.018)	(59.920)	(6.915)	(15.391)
Conservation	80.098***	−0.283	−2.097	36.723***	−3.550***	−10.238***
	(14.372)	(1.017)	(2.252)	(11.021)	(1.225)	(3.391)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	5076	5069	5076	5076	5069	5076

Standard errors in parentheses. Standard errors are clustered on crop districts, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

Conclusions

This article examined the claim that land use restrictions to protect endangered species impose a burden on farmers and hence reduce farm profits and land values. These land use restrictions are viewed as costly, so it should not be too surprising that the weight of evidence indicates that listing an endangered species reduces farm profits and land values. The preferred model, which controls for wide range of potential confounding factors by combining a matching design with a fixed effects panel data estimator, indicates that listing reduces land values by on average $131/acre, or 4% of current values, in dryland counties with habitat. There is little evidence that losses occur in irrigated counties or that losses are greater when a listing includes critical habitat.

These results have important policy implications. Attempts to reauthorize and amend the ESA have proposed full compensation to landowners if regulations reduce property values past a threshold, variously proposed between 20 and 33% of the pre-listing value. The economic implications of this policy aside, the results in this article show that the effect of listing on the margin does not come close to exceeding these levels, at least on average. It appears unlikely that much farmland would qualify for compensation unless landowners could recover the sum of their losses across multiple listings, although the majority of counties harbor fewer than two endangered species. Of course, the county-level effects reported here may mask heterogeneity at smaller spatial scales; some farmland may be more directly affected by land use restrictions than others in a county, depending on the location of habitat. In any case, the evidence in this article indicates that farmers are burdened but not devastated by ESA regulations.

References

Abadie, Alberto, and Guido W Imbens. 2008. On the Failure of the Bootstrap for

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Matching Estimators. *Econometrica* 76 (6): 1537–57.

AFBF. 2018. *Endangered Species Act Fact Sheet*, Washington, DC: American Farm Bureau Federation. https://www.fb.org/issues/regulatory-reform/the-endangered-species-act/.

Baker, Timothy G, Michael D Boehlje, and Michael R Langemeier. 2014. Farmland: Is it Currently Priced as an Attractive Investment? *American Journal of Agricultural Economics* 96(5): 1321–33.

Bošković, Branko, and Linda Nøstbakken. 2017. The Cost of Endangered Species Protection: Evidence from Auctions for Natural Resources. *Journal of Environmental Economics and Management* 81: 174–92.

Byl, Jacob P. 2019. Perverse Incentives and Safe Harbors in the Endangered Species Act: Evidence from Timber Harvests Near Woodpeckers. *Ecological Economics* 157: 100–8.

Cameron, A Colin, and Douglas L Miller. 2015. A Practitioner's Guide to Cluster-robust Inference. *Journal of Human Resources* 50(2): 317–72.

Copeland, Claudia (2010). Air Quality Issues and Animal Agriculture: A Primer. Washington DC: Congressional Research Service. http://nationalaglawcenter.org/wp-content/uploads/assets/crs/RL32948.pdf

Darwin, Roy. 1999. The Impact of Global Warming on Agriculture: A Ricardian Analysis: Comment. *American Economic Review* 89(4): 1049–52.

Davis, Charles. 2001. *Western Public Lands and Environmental Politics*. Boulder, CO: Westview Press.

Deschenes, Olivier, and Michael Greenstone. 2007. The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather. *American Economic Review* 97(1): 354–85.

Fallin, Mary, John Barrasso, and Virginia Foxx (2016). Republican Platform 2016. https://prod-cdn-static.gop.com/media/documents/DRAFT_12_FINAL[1]-ben_1468872234.pdf?_ga=2.157626994.933528283.1595281815-1680048605.1595281815

Ferraro, Paul J, and Jose J Miranda. 2017. Panel Data Designs and Estimators as Substitutes for Randomized Controlled Trials in the Evaluation of Public Programs. *Journal of the Association of Environmental and Resource Economists* 4(1): 281–317.

Fisher, Anthony C, W Michael Hanemann, Michael J Roberts, and Wolfram Schlenker. 2012. The Economic Impacts of Climate Change: Evidence from Agricultural Output and Random Fluctuations in Weather: Comment. *American Economic Review* 102(7): 3749–60.

FWS. 2019. Endangered and Threatened Wildlife and Plants; Regulations for Listing Species and Designating Critical Habitat. *Federal Registrar* 84(166): 45020–53.

GAO (1994). Endangered Species Act: Information on Species Protection on Nonfederal Lands. *U.S. General Accounting Office Report to Congressional Requesters* (GAO/RCED-95-16).

Geoghegan, Jacqueline, Lisa A Wainger, and Nancy E Bockstael. 1997. Spatial Landscape Indices in a Hedonic Framework: An Ecological Economics Analysis Using GIS. *Ecological Economics* 23(3): 251–64.

Goodstein, Eban. 1999. *The Trade-Off Myth: Fact and Fiction about Jobs and the Environment*. Washington DC: Island Press.

Greenstone, Michael, and Ted Gayer. 2009. Quasi-Experimental and Experimental Approaches to Environmental Economics. *Journal of Environmental Economics and Management* 57(1): 21–44.

Guiling, Pam, B Wade Brorsen, and Damona Doye. 2009. Effect of Urban Proximity on Agricultural Land Values. *Land Economics* 85(2): 252–64.

Guttman, Nathaniel B, and Robert G Quayle. 1996. A Historical Perspective of Us Climate Divisions. *Bulletin of the American Meteorological Society* 77(2): 293–303.

Imbens, Guido W, and Jeffrey M Wooldridge. 2009. Recent Developments in the Econometrics of Program Evaluation. *Journal of Economic Literature* 47(1): 5–86.

Innes, Robert, Stephen Polasky, and John Tschirhart. 1998. Takings, Compensation and Endangered Species Protection on Private Lands. *Journal of Economic Perspectives* 12(3): 35–52.

List, John A, Michael Margolis, and Daniel E. Osgood (2006). Is the Endangered Species Act Endangering Species? Technical report, National Bureau of Economic Research.

Lueck, Dean, and Jeffrey A Michael. 2003. Preemptive Habitat Destruction under the Endangered Species Act. *Journal of Law and Economics* 46(1): 27–60.

Malone, Trey, and Richard T Melstrom. 2020. Where's the Beef? Cattle Producers' Response to Endangered Species Listings. *Food Policy* 91: 101814.

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Melstrom, Richard T. 2017. Where to Drill? The Petroleum Industry's Response to an Endangered Species Listing. *Energy Economics* 66: 320–7.

———. 2019. A Note on the Petroleum Industry's Participation in Conservation Agreements for the Lesser Prairie Chicken. *Agricultural and Resource Economics Review* 48(1): 100–16.

Meltz, Robert. 2009. *The Endangered Species Act (ESA) and Claims of Property Rights "takings"*. Library of Congress: Congressional Research Service.

Mendelsohn, Robert, and Ariel Dinar. 2003. Climate, Water, and Agriculture. *Land Economics* 79(3): 328–41.

Mendelsohn, Robert, William D Nordhaus, and Daigee Shaw. 1994. The Impact of Global Warming on Agriculture: A Ricardian Analysis. *American Economic Review* 84(4): 753–71.

Meyer, Stephen M. 1997. The Economic Impact of the Endangered Species Act on the Housing and Real Estate Markets. *NYU Environmental Law Journal* 6: 450–79.

Michael, Jeffrey A. 2000. The Endangered Species Act and Private Landowner Incentives. In *Human Conflicts with Wildlife: Economic Considerations*, Vol 3. Fort Collins, Colorado: USDA National Wildlife Research Center Symposia.

NASS (2014). Farms and Farmland. *2012 Census of Agriculture Highlights* (ACH12-13). https://www.nass.usda.gov/Publications/Highlights/2014/Highlights_Farms_and_Farmland.pdf

Nelson, Erik J, John C Withey, Derric Pennington, and JJ Lawler. 2017. Identifying the Impacts of Critical Habitat Designation on Land Cover Change. *Resource and Energy Economics* 47: 89–125.

Njuki, Eric, and Boris E Bravo-Ureta. 2015. The Economic Costs of Environmental Regulation in US Dairy Farming: A Directional Distance Function Approach. *American Journal of Agricultural Economics* 97(4): 1087–106.

Plantinga, Andrew J, and Douglas J Miller. 2001. Agricultural Land Values and the Value of Rights to Future Land Development. *Land Economics* 77(1): 56–67.

Plantinga, Andrew J, Ted L Helvoigt, and Kirsten Walker. 2014. Critical Habitat for Threatened and Endangered Species: How Should the Economic Costs Be Evaluated? *Journal of Environmental Management* 134: 127–35.

Quigley, John M, and Aaron M Swoboda. 2007. The Urban Impacts of the Endangered Species Act: A General Equilibrium Analysis. *Journal of Urban Economics* 61(2): 299–318.

Ready, Riard C, and Charles W Abdalla. 2005. The Amenity and Disamenity Impacts of Agriculture: Estimates from a Hedonic Pricing Model. *American Journal of Agricultural Economics* 87(2): 314–26.

Roka, Fritz M, and Raymond B Palmquist. 1997. Examining the Use of National Databases in a Hedonic Analysis of Regional Farmland Values. *American Journal of Agricultural Economics* 79(5): 1651–6.

Rosenbaum, Paul R, and Donald B Rubin. 1985. Constructing a Control Group Using Multivariate Matched Sampling Methods that Incorporate the Propensity Score. *American Statistician* 39(1): 33–8.

Schlenker, Wolfram, W Michael Hanemann, and Anthony C Fisher. 2005. Will Us Agriculture Really Benefit from Global Warming? Accounting for Irrigation in the Hedonic Approach. *American Economic Review* 95(1): 395–406.

Sheridan, Thomas E. 2007. Embattled Ranchers, Endangered Species, and Urban Sprawl: The Political Ecology of the New American West. *Annual Review of Anthropology* 36: 121–38.

Srinivasan, U Thara, Susan P Carey, Eric Hallstein, Paul A Higgins, Amber C Kerr, Laura E Koteen, Adam B Smith, Reg Watson, John Harte, and Reg B Norgaard. 2008. The Debt of Nations and the Distribution of Ecological Impacts from Human Activities. *Proceedings of the National Academy of Sciences* 105(5): 1768–73.

Sunding, David, and David Zilberman. 2002. The Economics of Environmental Regulation by Licensing: An Assessment of Recent Changes to the Wetland Permitting Process. *Natural Resources Journal* 2(1): 59–90.

Swinton, Scott, Natalie Rector, G. Philip Robertson, Christina Jolejole-Foreman, and Frank Lupi. 2015. Farmer Decisions about Adopting Environmentally Beneficial Practices. In *The Ecology of Agricultural Landscapes: Long-Term Research on the Path to Sustainability*, ed. Stephan Hamilton, Julie Doll and G. Philip Robertson, 340–59. New York: Oxford.

Tegtmeier, Erin M, and Michael D Duffy. 2004. External Costs of Agricultural Production in the United States. *International*

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Journal of Agricultural Sustainability 2(1): 1–20.

Tilman, David, Joseph Fargione, Brian Wolff, Carla D'antonio, Andrew Dobson, Robert Howarth, David Schindler, William H Schlesinger, Daniel Simberloff, and Deborah Swackhamer. 2001. Forecasting Agriculturally Driven Global Environmental Change. *Science* 292(5515): 281–4.

Tscharntke, Teja, Alexandra M Klein, Andreas Kruess, Ingolf Steffan-Dewenter, and C Thies. 2005. Landscape Perspectives on Agricultural Intensification and Biodiversity–Ecosystem Service Management. *Ecology Letters* 8(8): 857–74.

Turner, Andrew J, and Kerry L McGrath. 2013. A Wider View of the Impacts of Critical Habitat Designation. *Environmental Law Reporter News & Analysis* 43: 10–678.

Wang, Yongji, Hongwei Cai, Chanjuan Li, Zhiwei Jiang, Ling Wang, Jiugang Song, and Jielai Xia. 2013. Optimal Caliper Width for Propensity Score Matching of Three Treatment Groups: A Monte Carlo Study. *PloS One* 8(12): e81045.

Wietelman, Derek C, and Richard T Melstrom. 2017. The Effect of Listing the Lesser Prairie Chicken as a Threatened Species on Rural Property Values. *Journal of Environmental Management* 191: 155–61.

Wilcove, David S, David Rothstein, Jason Dubow, Ali Phillips, and Elizabeth Losos. 1998. Quantifying Threats to Imperiled Species in the United States. *BioScience* 48(8): 607–15.

Wooldridge, Jeffrey M. 2002. *Econometric Analysis of Cross Section and Panel Data*. Cambridge, MA: MIT Press.

Zabel, Jeffrey E, and Robert W Paterson. 2006. The Effects of Critical Habitat Designation on Housing Supply: An Analysis of California Housing Construction Activity. *Journal of Regional Science* 46(1): 67–95.

Zhang, Daowei. 2004. Endangered Species and Timber Harvesting: The Case of Red-Cockaded Woodpeckers. *Economic Inquiry* 42(1): 150–65.

Zhang, Wei. 2017. Costs of a Practice-Based Air Quality Regulation: Dairy Farms in the San Joaquin Valley. *American Journal of Agricultural Economics* 100(3): 762–85.

Appendix

Additional Information about ESA Regulations

Prior to a 2019 rule change, the ESA conveyed the same protections from harm and habitat destruction for threatened and endangered species. ESA regulations should therefore affect land use in any area with the habitat of a listed species during the 2002 to 2017 study period, regardless of whether the FWS listed a species as threatened or endangered. To be clear, the expected cost of regulations can affect how the FWS tailors regulations to protect a species, which can be associated with listed status (Meltz 2009). For example, the ESA's 4(d) rule allows normal restrictions to be relaxed for threatened species if doing so will ease conflicts that would otherwise adversely affect recovery. However, it should be noted that the FWS can relax the take prohibition for both threatened and endangered species for certain land use activities through landowner agreements. Habitat conservation plans (HCPs) provide participating landowners an incidental take permit to use their land, as long as they agree to minimize and mitigate the effects of the authorized incidental take. HCPs were created in response to concerns that ESA regulations created perverse incentives for landowners to avoid land use restrictions by destroying habitat. Safe harbor agreements (SHAs) provide regulatory assurances to participating landowners that they will not be subject to additional restrictions if their land use activities improve species habitat. Candidate conservation agreements with assurances (CCAAs) are a pre-listing tool that provides assurances to participating landowners that if they complete certain conservation activities they will not be subject to additional restrictions if the species is listed in the future.

This study analyzes the effect of endangered and threatened species on agriculture without conditioning on participation in HCPs, SHAs and CCAAs (although the analysis does include a variable for land enrolled in conservation easements, which could correlate with participation in HCPs, SHAs and CCAAs). Recent research has found participation in voluntary conservation agreements to be low and the effects on behavior modest (Byl 2019; Melstrom 2019), which implies that the results are not likely to change

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

substantially after conditioning on participation in ESA conservation agreements.

Comparing Land Values in the Census and Prices from Actual Sales

As a check on the accuracy of the census estimates, I collected data on agricultural land sales in Minnesota and Oklahoma. The data from Minnesota are publically available at http://www.landeconomics.umn.edu/. The Oklahoma data come from a previously published study on land prices and ESA regulations, which overlap with the 2007 and 2012 censuses (Wietelman and Melstrom 2017). Following the practice of earlier hedonic studies (Guiling, Brorsen, and Doye 2009), I filtered out parcels with a price less than $150 per acre (in 2017$) to exclude parcels transferred between family members, or greater than $10,000 per acre (in 2017$) to exclude residential parcels mistakenly classified as agricultural. I then calculated the parcel size-weighted price per acre in each county in each census year. Table A1 summarizes these prices and the census estimates, in nominal values, after excluding counties with more than 10% of land classified as urban. The two series closely track each other. Although there is a statistically significant difference in Minnesota in 2012, the census estimates and sales prices are generally similar across counties and years, with an overall correlation coefficient of 0.96. Note that Roka and Palmquist (1997) find no evidence that farmers systematically over or understate land values on USDA surveys. Specifically, they find that farmers who recently bought or sold land reported land values that were statistically indistinguishable from the values reported by other farmers.

NatureServe Element Occurrence Scores

Element occurrence scores are a measure of the probability that a population of a species will persist in an area. NatureServe assigns scores based on reports from a network of state natural heritage programs and some state natural resource agencies, which regularly track the locations of listed species within their jurisdictions. Updates occur every twelve to eighteen months. Sometimes there are separate populations within a county, which may have different scores; in these cases, the county score is the highest observed score.

The set of scores is A (excellent), B (good), C (marginal), D (poor), E (verified extant but not assessed), F (failed to find), H (historical), X (extirpated), NR (not ranked), and U (unranked). Use of the E rank includes, in general, locations where the species is thought to be extant, but an A, B, C or D rank cannot be assigned. Scores A–E therefore indicate that the species is present. A score of F means that the species was not found despite a search by experienced observers at a location where it was previously reported, so it is possible the species has been extirpated. A score of H includes occurrences based on historical collection data, or occurrences that previously were ranked A, B, C, D, or E but need new survey work due to habitat loss. A score of X indicates that surveys by experienced observers show that the species no longer exists or that habitat has been destroyed to an extent that the area can no longer support the species. For further information, see http://explorer.natureserve.org/eorankguide.htm.

In the analysis, I assume a species is present if it has a score of A, B, C, D, E, or H in a county. The results are qualitatively similar if the analysis excludes H or includes F when

Table A1. Census Land Values versus Land Prices in Minnesota and Oklahoma

	Ag census		Land sales		
	Mean	St. err.	Mean	St. err.	Observations
Minnesota					
2002	1526	70	1588	95	76
2007	2616	89	2700	128	76
2012	4050	190	4466	251	76
2017	4635	206	4460	222	76
Oklahoma					
2007	1228	45	1351	88	68
2012	1460	51	1493	52	73

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A2. Number of New Endangered Species by 5-year Period

Taxon	1982	1987	1992	1997	2002	2007	2012	2017
Amphibians	7	1	2	1	5	2	2	8
Birds	42	5	6	6	2	0	0	3
Flowering plants	49	72	144	286	133	8	43	102
Freshwater mussels	19	1	17	14	13	2	2	16
Freshwater fishes	35	29	19	17	25	16	7	8
Mammals	18	12	14	3	9	3	3	3
Terrestrial snails	27	0	3	1	1	0	0	2
Other	27	9	19	36	36	14	11	36
Total	224	129	234	364	224	45	68	178

Table A3. Coefficient Estimates Excluding Counties with More than Ten Endangered Species

	All counties			Dryland counties		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−118.702***	−0.757	−4.214	−151.983***	−3.217	−9.314
	(40.042)	(2.242)	(5.390)	(39.941)	(2.588)	(5.759)
Operator age	1419.914***	117.464***	395.123***	2058.475***	165.752***	484.380***
	(355.282)	(24.178)	(67.549)	(467.435)	(36.234)	(86.150)
Operator age^2	−11.838***	−0.996***	−3.362***	−17.259***	−1.414***	−4.132***
	(3.023)	(0.206)	(0.578)	(3.995)	(0.312)	(0.738)
Operator female	−4.522**	−0.029	−0.206	−4.928	−0.013	−0.068
	(1.958)	(0.131)	(0.304)	(3.359)	(0.212)	(0.463)
Income	−7.247	−0.908***	−2.468***	−8.317	−1.051**	−3.011***
	(6.527)	(0.326)	(0.807)	(8.040)	(0.484)	(1.083)
Population/sqmi	21.132***	−0.641***	0.925*	16.856***	−0.682***	−0.234
	(4.224)	(0.201)	(0.516)	(4.324)	(0.256)	(0.576)
Crops	16.808***	2.005***	5.086***	22.398***	2.705***	5.871***
	(5.904)	(0.353)	(0.954)	(6.418)	(0.475)	(1.098)
Irrigation	38.412**	0.743	7.808***	156.480***	10.426**	28.815***
	(15.742)	(0.905)	(2.486)	(59.337)	(4.935)	(10.785)
Conservation	31.066***	−0.488	−3.387**	50.672***	−1.365	−4.614**
	(8.299)	(0.652)	(1.386)	(12.714)	(0.926)	(2.291)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	8658	8650	8658	5008	5001	5008

Standard errors in parentheses. Standard errors are clustered on climate divisions, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

determining presence. Later in the Appendix I present results when the number of endangered species in a county is based on FWS rather than NatureServe habitat data.

Results with Different Research Designs and Types of Species

Table A2 shows that the FWS listed 291 species as endangered or threatened between 2002 and 2017. This list does not, in general, include marine species listed by the NMFS. Flowering plants compose half of listings since 2002. The next largest category, labeled "other," includes reptiles, crayfishes, freshwater snails, insects, conifers, lichens, moss, and ferns, among other species. In the main text, endangered plants includes flowering plants, conifers, lichens, moss, and ferns. Later in this section, I present separate effects for aquatic and nonaquatic animals species. Aquatic animals include

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A4. Coefficient Estimates for Dryland Counties with Smaller Sample and Separate Counts for Animal Types

	Smaller sample			Aquatic vs. nonaquatic animals		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−41.267*	−5.388*	−10.781**			
	(22.731)	(2.763)	(4.971)			
Number of nonaquatic ES				−82.937	−4.085	1.119
				(65.410)	(3.963)	(8.846)
Number of aquatic ES				−160.348***	−5.594*	−19.211***
				(43.957)	(3.310)	(6.063)
Operator age	155.893	48.264**	218.796***	2056.080***	164.497***	480.291***
	(344.527)	(23.090)	(57.666)	(466.960)	(36.022)	(85.947)
Operator age^2	−1.511	−0.411**	−1.930***	−17.239***	−1.404***	−4.099***
	(2.969)	(0.201)	(0.502)	(3.990)	(0.310)	(0.736)
Operator female	1.709	−0.577	−1.055	−4.946	−0.015	−0.058
	(3.583)	(0.421)	(0.933)	(3.133)	(0.206)	(0.436)
Income	1.742	−0.375	−2.037**	−8.165	−1.069**	−3.045***
	(4.686)	(0.466)	(0.935)	(8.008)	(0.483)	(1.081)
Population/sqmi	9.502***	−0.618*	0.331	16.938***	−0.667***	−0.173
	(2.827)	(0.350)	(0.822)	(4.259)	(0.249)	(0.565)
Crops	17.333***	2.402***	5.605***	22.607***	2.740***	5.958***
	(4.163)	(0.378)	(0.875)	(6.387)	(0.474)	(1.096)
Irrigation	26.830	7.303	20.301**	153.961***	10.433**	28.947***
	(49.002)	(5.705)	(9.389)	(58.657)	(4.905)	(10.769)
Conservation	1.433	−4.172***	−9.690***	50.569***	−1.322	−4.440**
	(7.917)	(1.063)	(2.922)	(12.651)	(0.915)	(2.249)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	3068	3062	3068	5076	5069	5076

Standard errors in parentheses. Standard errors are clustered on climate divisions, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

freshwater mussels, fish, snails, and crayfish. Nonaquatic animals includes all other animal species, including mammals, birds, reptiles, and amphibians. Roughly half of endangered animals are aquatic under these definitions.

Although most counties contain less than two endangered species, about 5% contain more than ten. The results in the main text show that the effect of listing an endangered species on land value, profit and revenue is negative in dryland counties. Are these results sensitive to omitting counties with large numbers of endangered species? Table A3 provides evidence that the effect of endangered species on land value, profit, and revenue becomes somewhat more negative after dropping counties with more than ten endangered species. Columns (1) to (3) show the coefficients when irrigated and dryland counties are pooled, and columns (4) to (6) show the coefficients for dryland counties only. The most notable difference between the results in table A3 and table 3 in the main text is that the coefficient on the number of endangered species in the land value model decreases from \$-81 to \$-119.

Table A4, columns (1) to (3) report the coefficients when the models are estimated on a small but more homogenous group of counties. I limited the data to dryland counties with between 25% and 75% of farmland in crops, which is approximately 1 standard deviation below and above the average percent of farmland in crops. This removes counties that are farmed predominantly as pasture or cropland. The coefficient on the number of endangered species in the land value model is negative but not significantly differently from zero. Compared with the results reported in table 4, this change could reflect real

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A5. Coefficient Estimates for Dryland Counties Using Alternative Source of Habitat Data

	Location of ES based on FWS habitat			Instrumenting FWS habitat		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−83.772***	−5.283**	−5.767	−168.960***	−5.666*	−12.795*
	(21.632)	(2.265)	(3.920)	(46.659)	(3.211)	(6.789)
Operator age	2042.082***	163.185***	480.735***	2015.403***	163.066***	478.534***
	(456.955)	(35.422)	(85.217)	(448.599)	(35.349)	(84.045)
Operator age^2	−17.095***	−1.392***	−4.101***	−16.858***	−1.391***	−4.081***
	(3.906)	(0.305)	(0.730)	(3.837)	(0.304)	(0.719)
Operator female	−4.673	−0.007	−0.038	−4.808	−0.007	−0.050
	(3.203)	(0.211)	(0.449)	(3.407)	(0.211)	(0.464)
Income	−7.479	−1.043**	−2.990***	−7.544	−1.043**	−2.996***
	(7.991)	(0.480)	(1.081)	(7.913)	(0.480)	(1.076)
Population/sqmi	16.110***	−0.720***	−0.241	15.187***	−0.724***	−0.317
	(4.527)	(0.259)	(0.583)	(4.742)	(0.260)	(0.596)
Crops	22.706***	2.738***	5.973***	22.481***	2.737***	5.955***
	(6.325)	(0.469)	(1.095)	(6.303)	(0.470)	(1.092)
Irrigation	149.871**	10.272**	28.546***	149.490**	10.271**	28.515***
	(59.238)	(4.935)	(10.798)	(61.209)	(4.937)	(10.918)
Conservation	51.754***	−1.193	−4.476*	54.816***	−1.180	−4.223*
	(12.908)	(0.931)	(2.292)	(13.423)	(0.949)	(2.356)
Soil and weather controls, county and division-by-year effects	Yes	Yes	Yes	Yes	Yes	Yes
N	5076	5069	5076	5076	5069	5076

Standard errors in parentheses. Standard errors are clustered on climate divisions, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

differences in the effect of ESA regulations based on the type of farming. However, the change could also be spurious, because the coefficients on the number of endangered species in the profit and revenue models are both negative, statistically significant, and the same magnitude as their counterparts in table 4.

Table A4, columns (4) to (6) report the coefficients when the models include separate variables for aquatic and nonaquatic animals. The point estimates associated with the effect of listing nonaquatic animals are closer to zero than the same effect for aquatic animals. However, these effects are estimated imprecisely. Note that the coefficients on the number of listed nonaquatic animals have standard errors around 1.5 times the size the same standard errors in tables 4. A test of coefficient equality rejects the hypothesis that listing aquatic and nonaquatic animals has the same effect on revenue ($p < 0.05$). Equality cannot be rejected in the case of land value ($p = 0.30$) or profit ($p = 0.78$).

Table A5, columns (1) to (3) report the coefficients using different habitat data. Like NatureServe, the FWS reports the counties that listed species are known or believed to be present. In general, FWS habitat data are much more diffuse and imprecise than NatureServe habitat data—for example, FWS habitat data are defined at the state level rather than the county level for some species—such that the number of endangered species in a county is about five on average based on the FWS data, which is more than twice the number based on the NatureServe data. However, there is a high correlation between the two numbers ($\rho = 0.78$). Columns (1) to (3) report the results for dryland counties when the number of endangered species in a county is determined by the FWS data. Compared with the results in the main text, the coefficient on the number of endangered species is less negative in the land value model and about the same in the profit and revenue models.

Now consider the issue of measurement error in the endangered species variable, which is the difference between the actual number of endangered species in a county

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Table A6. Coefficient Estimates for Dryland Counties with Different Varieties of Fixed Effects

	County and year FEs			County and state-by-year FEs		
	(1) Land value	(2) Profit	(3) Revenue	(4) Land value	(5) Profit	(6) Revenue
Number of ES	−170.200***	−5.098**	−13.643***	−64.259**	−1.805	−6.577
	(41.893)	(2.189)	(5.180)	(28.675)	(1.894)	(4.151)
Operator age	3276.043***	214.248***	621.388***	1539.974***	130.545***	364.754***
	(561.394)	(45.723)	(102.880)	(377.864)	(31.594)	(72.621)
Operator age^2	−27.662***	−1.860***	−5.355***	−12.960***	−1.115***	−3.130***
	(4.799)	(0.395)	(0.884)	(3.224)	(0.272)	(0.626)
Operator female	−10.544**	0.007	−0.648	−4.830*	0.028	−0.303
	(4.760)	(0.216)	(0.569)	(2.778)	(0.196)	(0.416)
Income	−5.995	−1.657***	−4.126***	−6.115	−0.494	−0.880
	(8.416)	(0.466)	(1.169)	(5.346)	(0.480)	(0.988)
Population/sqmi	14.891***	−0.655**	−0.191	18.115***	−0.692***	−0.314
	(4.993)	(0.262)	(0.583)	(4.712)	(0.221)	(0.480)
Crops	33.479***	3.214***	6.734***	13.112***	2.272***	4.722***
	(7.012)	(0.510)	(1.087)	(4.174)	(0.443)	(0.869)
Irrigation	220.724***	6.465	26.724**	121.438***	8.190*	25.824***
	(70.436)	(5.199)	(11.259)	(41.804)	(4.454)	(7.929)
Conservation	44.077**	−1.473*	−4.596*	34.288***	0.161	−2.154
	(18.310)	(0.881)	(2.644)	(11.879)	(0.896)	(1.695)
Soil and weather controls	Yes	Yes	Yes	Yes	Yes	Yes
N	5076	5069	5076	5076	5069	5076

Standard errors in parentheses. Standard errors are clustered on climate divisions, allowing for correlations across groups of counties within states.
*$p < 0.10$.
**$p < 0.05$.
***$p < 0.01$.

and the number reported in the data. With measurement error, under the classical errors-in-variables assumption, the coefficient estimates will suffer from attenuation bias. Thus, the accuracy of the NatureServe and FWS habitat data has a significant effect on the amount of measurement error bias in the models. Because these two sources of habitat data do not produce identical numbers, clearly at least one must be measured with error. However, having two measures, we can use one in the model and use the other as an instrument, which will identify the true coefficient (Wooldridge 2002). I find that when I use the number of endangered species based on the NatureServe data as an instrument for the number of endangered species based on the FWS data, the first-stage *F*-statistic for instrument significance exceeds 100, which is far above the rule of thumb that the statistic should be larger than 10. Columns (4) to (6) report the results having instrumented. These results indicate that a listing on average reduces land values \$69 ($p < 0.01$), profit \$6 ($p < 0.10$) and revenue \$13 ($p < 0.10$) on average in counties with habitat. Thus, the estimates in table A5, columns (4) to (6) imply that the results in the main text are not significantly biased due to measurement error.

Results with Different Varieties of Fixed Effects

Including census division-by-year fixed effects reduces the susceptibility of the model to bias from region-specific shocks that could correlate with listing. If there is measurement error, however, adding fixed effects will amplify attenuation bias. Fisher et al. (2012) show that state-by-year effects in hedonic models of county-level agricultural census data absorb a significant amount of weather variation and thus amplify attenuation bias in the weather variable coefficients. Fixed effects may also amplify attenuation bias in the coefficient of endangered species variable, either because the variable itself is measured with error or because it is correlated with other variables measured with error (in fact, listings are significantly

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

correlated with weather variation, based on a regression of listings on the weather variables, county and division-by-year fixed effects). Thus, if richer varieties of fixed effects absorb useful variation in the data, bias amplification from measurement error could outweigh bias reduction from controlling for more omitted variables. It is therefore important to probe the sensitivity of hedonic estimates to various levels of fixed effects.

Table A6 presents the coefficients for varieties of fixed effects not shown in the main text. Across models, the coefficient on the number of endangered species diminishes as various levels of fixed effects are added, which is consistent with an increased influence of measurement error. Furthermore, the estimate from the models with state-by-year fixed effects move away from zero after instrumenting the number of endangered species based on the NatureServe data with the number based on the FWS data, which is consistent with correcting for measurement error bias. For example, the effect of listing an endangered species on land value is -$64/acre in the model with state-by-year effects (column 4), but the same effect after instrumenting the number of endangered species with the FWS locations is -$173/acre (results not shown for brevity). Regardless, as shown in table A6, even without correcting for potential measurement error bias, when the specification includes county and state-by-year effects, the effect of endangered species is negative in all three models and significantly different from zero in the land value model ($p < 0.05$).

14678276, 2021, 1, Downloaded from https://onlinelibrary.wiley.com/doi/10.1111/ajae.12127 by Loyola University Chicago, Wiley Online Library on [26/04/2025]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License